

Multiple Choice

1. **Answer:** A

Options: A) 32; B) 45; C) 40; D) 49; E) 63

Note: Correct option: A - 32

2. **Answer:** D

Options: A) $3 + 393$; B) 3×393 ; C) $(393 - 3) \times 3$; D) $393 \div 3$; E) $(393 + 3) \div 3$

Note: Correct option: D - $393 \div 3$

3. **Answer:** A

Options: A) 12; B) 15; C) 14; D) 16; E) 22

Note: Correct option: A - 12

4. **Answer:** D

Options: A) $2\sqrt{3}$; B) $\sqrt{7}$; C) $3\sqrt{2}$; D) $5\sqrt{2}$; E) 2

Note: Correct option: D - $5\sqrt{2}$

5. **Answer:** A

Options: A) 0; B) 720; C) 30; D) 180; E) 90

Note: Correct option: A - 0

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '10'.

8. **Answer:** False

Options: A) True; B) False

Note: Actual answer is '5'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $p = 24$ '.

Long Form Response

11. **Answer:** Recall the formula $A = P \left(1 + \frac{r}{n}\right)^{nt}$, where A is the end balance, P is the principal, r is the interest rate, t is the number of years, and n is the number of times the interest is compounded in a year. This formula represents the idea that the interest is compounded every $1/n$ years with the rate of r/n . Substituting the given information, we have

$$60,000 = P \left(1 + \frac{0.07}{4}\right)^{4 \cdot 5}.$$

Solving for P gives $P = 42409.474\dots$, which rounded to the nearest dollar is \$42409}\$.*Note: Studentssh*

11. **Answer:** 181 feet 4 inches. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.